

# Rockchip RV Series Linux Audio Codec Developer Guide

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**Preface**

**Overview**

This document primarily introduces the features and functions of RV1106/RV1103 Audio Codec, as well as common attribute configurations of ACodec.

**Product Version**

| Chipset          | Kernel Version |
|------------------|----------------|
| RV1106/RV1103    | 5.10           |
| RV1106B/RV1103B  | 5.10           |
| RV1126B/RV1126BP | 6.1            |

**Intended Audience**

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

**Revision History**

| Version | Author     | Date       | Change Description                                                                                  |
|---------|------------|------------|-----------------------------------------------------------------------------------------------------|
| V1.0.0  | Xing Zheng | 2022-05-12 | Added the initial version of RV1106 Linux Audio Codec Developer Guide                               |
| V1.1.0  | Xing Zheng | 2022-08-31 | Added commonly used playback and recording test commands of RV1106 Audio Codec                      |
| V1.2.0  | Xing Zheng | 2023-10-20 | Fixed the range description error of ADC ALC node, added "DAC Control Manually" description         |
| V1.3.0  | Xing Zheng | 2023-12-18 | Fixed the range description error of ADC ALC node, and fiexed the reference of the invalid document |
| V1.4.0  | Xing Zheng | 2024-07-07 | Added the the introduction of RV1106B/RV1103B Audio Codec                                           |
| V1.5.0  | Xing Zheng | 2024-09-20 | Rename the document and unify the naming style                                                      |

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# 1. Overview

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## 1.1 Function Introduction

The ACodec mentioned here is the audio IP module built into the SoCs of RV1106/RV1103/RV1106B/RV1103B. It can be used to connect an external analog microphone and transmit the external signal recording to the CPU through analog-to-digital conversion. The CPU can also convert the local PCM audio digital signal through digital-to-analog conversion and then play it out.

The ACodec IP contained in RV1106 and RV1103 is exactly the same, including 2 ADCs and 1 DAC. The RV1106B/RV1103B SoC was launched in 2024, and the internal ACodec IP is exactly the same. Compared with RV1106/RV1103, its ACodec has deleted one ADC and is 1 ADC and 1 DAC. When operating ACodec, you will usually encounter the scenario of retrieving channel configuration, which will be explained in detail later in the document.

The RV1126B/RV1126BP, released in 2025, supports two low-power, high-performance ADCs, making them suitable for single- and dual-microphone voice pickup scenarios. ADC0 supports AOA (Always On Audio) technology, enabling ultra-low-power continuous recording and wake-up in sleep mode. For detailed instructions, refer to the AOA module section in the SDK documentation, "Rockchip Developer Guide Linux AOV CN.pdf" or "Rockchip Developer Guide Linux AOV EN.pdf."

## 2. ACodec Introduction

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### 2.1 ACodec Features

The basic features of RV1106/RV1103/RV1106B/RV1103B ACodec are as follows:

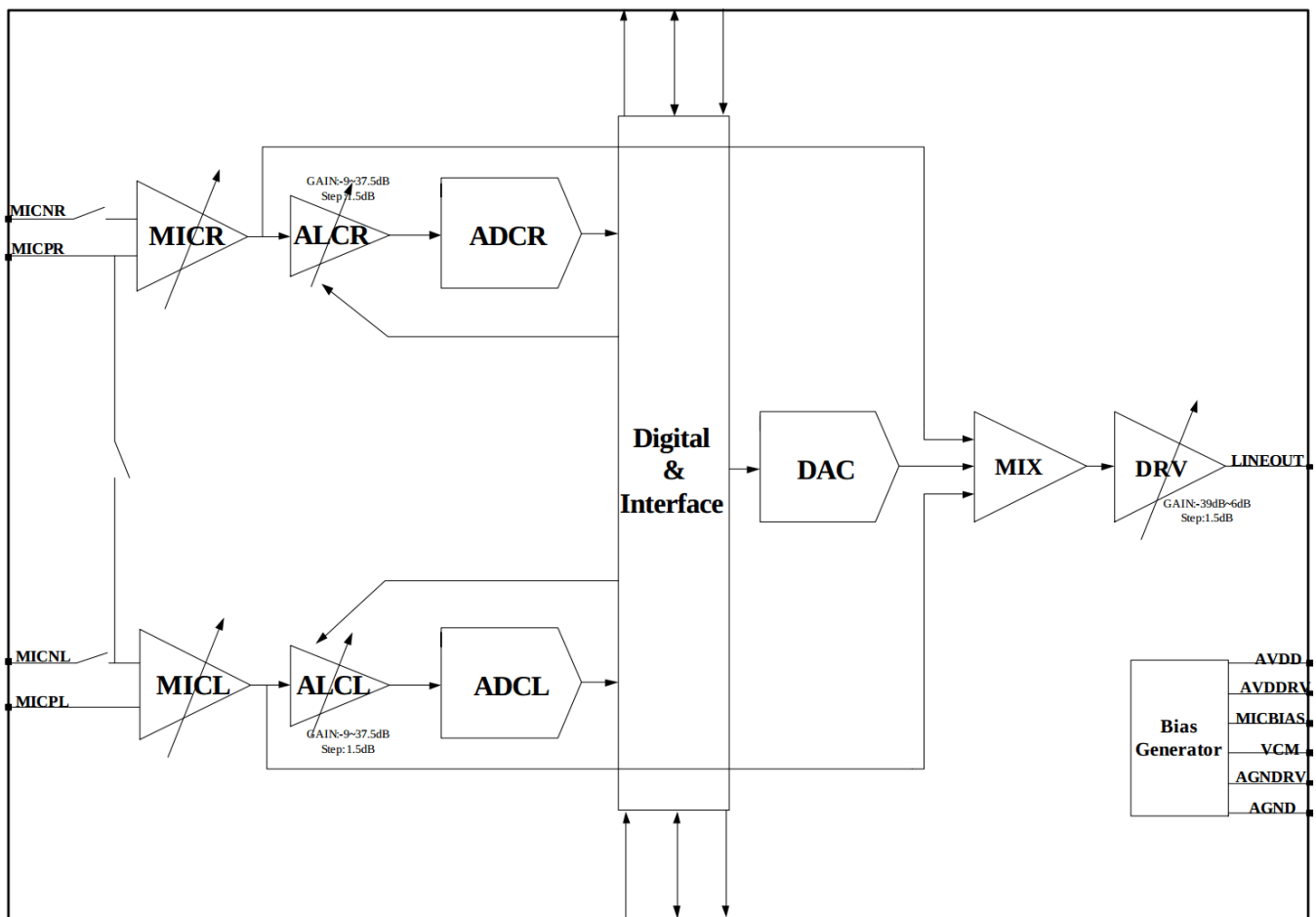
- 24 bits DAC with 93dB(A-weighted) SNR
- Support 600Ω line output
- Low power: 2.5mA for playback
- 24 bits ADC with 92dB(A-weighted) SNR
- Support differential and single-ended microphone or line input
- Low power: 5mA for stereo recording
- Automatic Level Control (ALC) for smooth audio recording
- Low power: less than 0.05mA for standby
- Programmable input and output analog gains
- Digital interpolation and decimation filter integrated
- Sampling rate of 8kHz/12kHz/16kHz/24kHz/32kHz/44.1kHz/48kHz/96kHz

- 1.8V supply for analog and 0.9V supply for digital

The basic features of the RV1126B/RV1126BP ACodec are as follows:

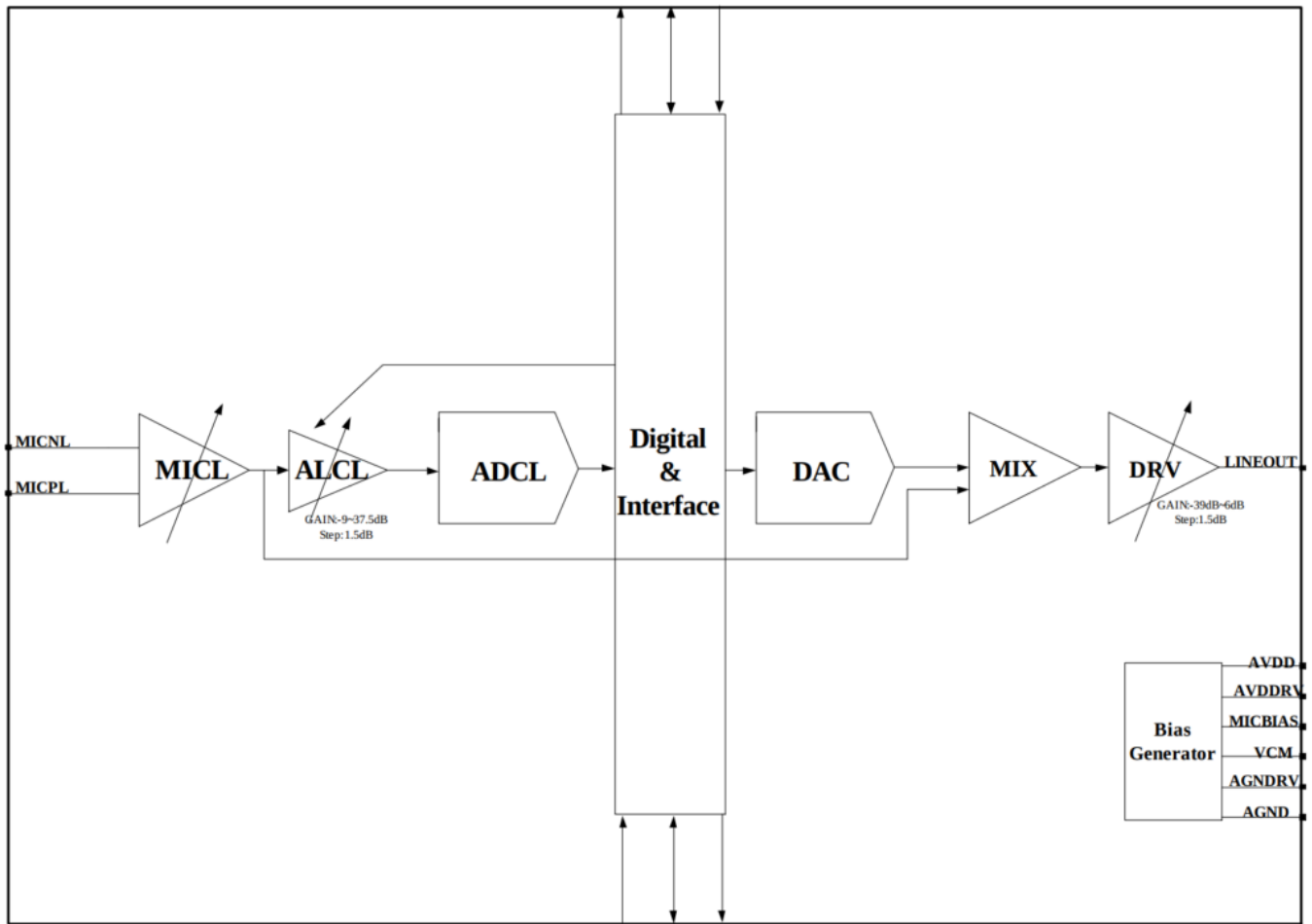
- Support APB bus slave interface
- Support Analog gain between 0~48dB with 3dB/step
- Support 96dB SNR
- Support 1-channel Differential-mode/Single-mode/Pseudo-differential mode ADC
- Support I2S/PCM master and slave mode
- Support I2S normal, left and right justified mode serial audio data transfer
- Support PCM early, late1, late2, late3 mode serial audio data transfer
- Support MSB or LSB first serial audio data transfer
- Support 16~32bit sample resolution
- Support sample rate: 8kHz, 16kHz, 32kHz, 64kHz, 128kHz, 11.025kHz, 22.05kHz, 44.1kHz, 88.2kHz, 176.4kHz, 12kHz, 24kHz, 48kHz, 96kHz, 192kHz
- The passband of filters is  $0.45625 * FS$
- Support passband ripple within  $\pm 0.1\text{dB}$
- The stopband of filters is  $0.5 * FS$
- Support stopband attenuation at least 60dB
- Support Digital Audio ADC volume control
- Support programmable negative volume gain and positive volume gain

## 2.2 ACodec Block Diagram



RV1106/RV1103 ACodec Block Diagram

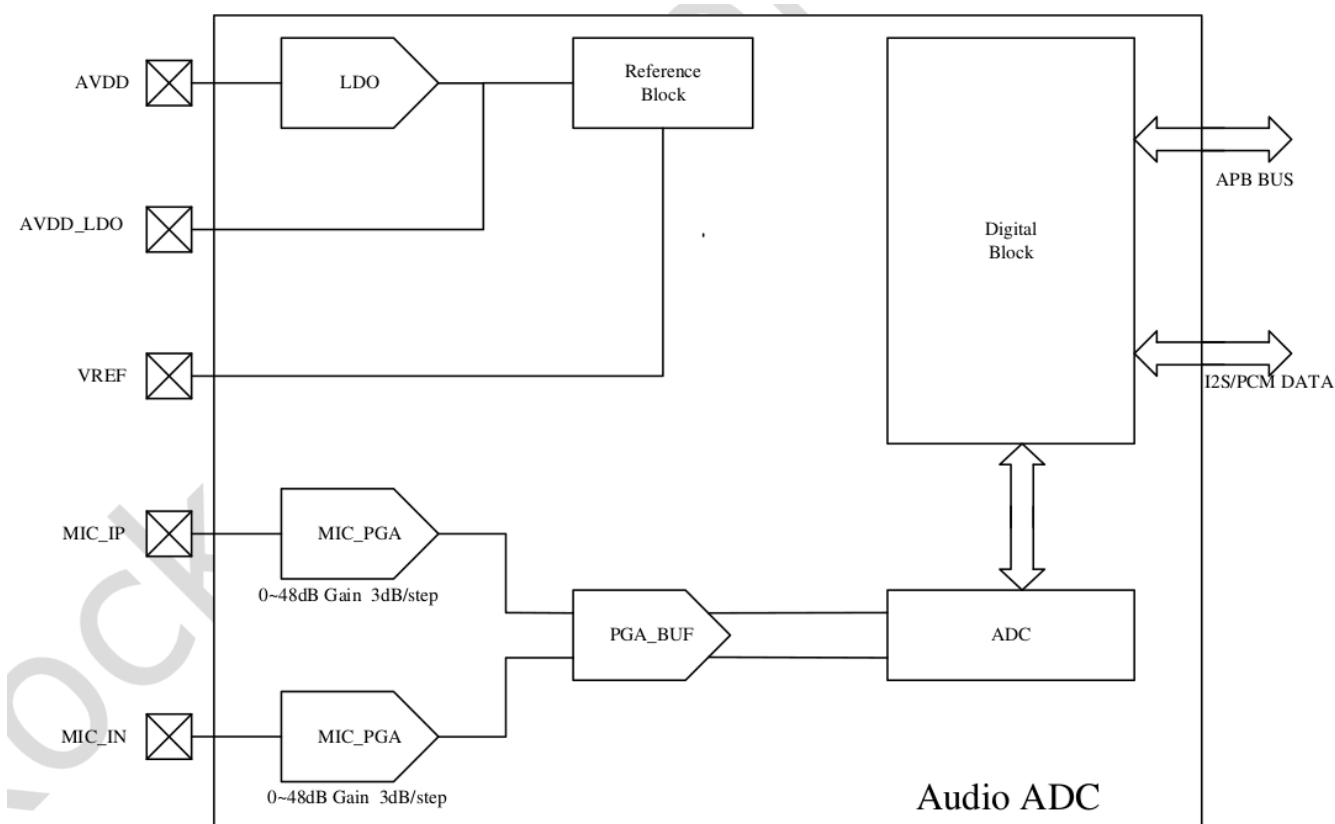
From the block diagram, it can be found that both RV1106 and RV1103 support 2 ADC inputs (for recording) and 1 DAC output (for playback). It's important to note that in differential mode, RV1106 can support a maximum of 2 microphones, while RV1103, in differential mode, only supports the connection of 1 microphone. In single-ended mode, both RV1106 and RV1103 can support a maximum of 2 microphones each.



RV1106B/RV1103B ACodec Block Diagram

The RV1106B/RV1103B ACodec removes one ADC in terms of function, and supports one ADC input (capture) and one DAC output (playback). However, the ADC input still maintains the two access modes of differential input or single-ended input.

The RV1126B/RV1126BP supports two low-power, high-performance ADCs by default, and both support differential inputs by default. The block diagram is as follows:



RV1126B/RV1126BP ACodec Block Diagram

Therefore, according to the SoC ID, the main differences of the following ACodec MIC access are:

| SoC ID           | Maximum number of MICs in single-ended mode | Maximum number of MICs in differential mode |
|------------------|---------------------------------------------|---------------------------------------------|
| RV1106           | 2                                           | 2                                           |
| RV1103           | 2                                           | 1                                           |
| RV1106B/RV1103B  | 1                                           | 1                                           |
| RV1126B/RV1126BP | 2                                           | 2                                           |

For typical MIC connection methods, please refer to RK EVB drawings and hardware design reference guide.

## 3. ACodec Software Configuration

### 3.1 RV1106/RV1103 ACodec Kernel Configuration

Taking Rockchip kernel-5.10 SDK code as an example, you can refer to the following dtsti file:

```
arch/arm/boot/dts/rv1106-evb-v10.dtsi
```



Firstly, define an `acodec_sound` node, which describes some basic information of the sound card and references the DAI (Digital Audio Interface) `i2s0_8ch` on the controller side and the DAI `acodec` on the peripheral side:

```
acodec_sound: acodec-sound {
    compatible = "simple-audio-card";
    simple-audio-card,name = "rv1106-acodec";
    simple-audio-card,format = "i2s";
    simple-audio-card,mclk-fs = <256>;
    simple-audio-card,cpu {
        sound-dai = <&i2s0_8ch>;
    };
    simple-audio-card,codec {
        sound-dai = <&acodec>;
    };
};
```

Enable I2S0:

```
&i2s0_8ch {
    #sound-dai-cells = <0>;
    status = "okay";
};
```

Enable ACodec and define the GPIO pin that controls the external power amplifier:

```
&acodec {
    #sound-dai-cells = <0>;
    pa-ctl-gpios = <&gpio1 RK_PA1 GPIO_ACTIVE_HIGH>;
    status = "okay";
};
```

In the above example, the I2S0 inside RV1106/RV1103 can be bound to ACodec.

**It should be noted that there is only one I2S controller inside RV1106/RV1103, which is I2S0. If the internal ACodec is enabled, I2S0 will be occupied at this time, and it is connected to ACodec through the SoC internal lead wires. Therefore, in this scenario, the pins connected to external IO by I2S0 will be unavailable.**

## 3.2 RV1106B/RV1103B ACodec Kernel Configuration

The configuration of RV1106B/RV1103B ACodec can refer to

```
arch/arm/boot/dts/rv1103b-evb.dtsi
```

```

acodec_sound: acodec-sound {
    compatible = "simple-audio-card";
    simple-audio-card,name = "rv1103b-acodec";
    simple-audio-card,format = "i2s";
    simple-audio-card,mclk-fs = <256>;
    simple-audio-card,cpu {
        sound-dai = <&sai>;
    };
    simple-audio-card,codec {
        sound-dai = <&acodec>;
    };
};

```

It can be seen that the digital audio interface controller of RV1106B/RV1103B has been upgraded from I2S to SAI, which needs to be enabled:

```

&sai {
    status = "okay";
};

```

You also need to enable the acodec node:

```

&acodec {
    pa-ctl-gpios = <&gpio0 RK_PA3 GPIO_ACTIVE_HIGH>;
    status = "okay";
};

```

### 3.3 RV1106/RV1103 ACodec User Level Configuration

Usually, we can use the common and authoritative tinymix tool to obtain and set the control node information that the codec exposes to user layer. However, since tinymix tool was cropped in the SDK, rockit integrates the rk\_mpi\_amix\_test tool that is equivalent to tinymix and can be used to list all contents information:

```

# rk_mpi_amix_test --list_contents
cmd parse result:
sound control id      : 0
control name          : (null)
control value         : (null)
list controls         : 0
list contents         : 1
Number of controls: 26

```

| ctl  | type | num | name                         | value                     |
|------|------|-----|------------------------------|---------------------------|
| 0    | ENUM | 1   | I2STDM Digital Loopback Mode | , DisabledMode1Mode2Mode2 |
| Swap |      |     |                              |                           |
| 1    | INT  | 1   | ADC MIC Left Gain            | 2 (range 0->3)            |
| 2    | INT  | 1   | ADC MIC Right Gain           | 2 (range 0->3)            |
| 3    | INT  | 1   | ADC ALC Left Volume          | 6 (range 0->31)           |
| 4    | INT  | 1   | ADC ALC Right Volume         | 6 (range 0->31)           |
| 5    | INT  | 1   | ADC Digital Left Volume      | 195 (range 0->255)        |

|    |      |   |                                   |                                                                                  |
|----|------|---|-----------------------------------|----------------------------------------------------------------------------------|
| 6  | INT  | 1 | ADC Digital Right Volume          | 195 (range 0->255)                                                               |
| 7  | ENUM | 1 | ADC HPF Cut-off                   | , OffOn                                                                          |
| 8  | INT  | 1 | ALC AGC Left Volume               | 12 (range 0->31)                                                                 |
| 9  | INT  | 1 | ALC AGC Right Volume              | 12 (range 0->31)                                                                 |
| 10 | INT  | 1 | ALC AGC Left Max Volume           | 7 (range 0->7)                                                                   |
| 11 | INT  | 1 | ALC AGC Right Max Volume          | 7 (range 0->7)                                                                   |
| 12 | INT  | 1 | ALC AGC Left Min Volume           | 0 (range 0->7)                                                                   |
| 13 | INT  | 1 | ALC AGC Right Min Volume          | 0 (range 0->7)                                                                   |
| 14 | ENUM | 1 | ALC AGC Left Switch               | , OffOn                                                                          |
| 15 | ENUM | 1 | ALC AGC Right Switch              | , OffOn                                                                          |
| 16 | ENUM | 1 | AGC Left Approximate Sample Rate  | ,<br>96KHz48KHz44.1KHz32KHz24KHz16KHz12KHz8KHz                                   |
| 17 | ENUM | 1 | AGC Right Approximate Sample Rate | ,<br>96KHz48KHz44.1KHz32KHz24KHz16KHz12KHz8KHz                                   |
| 18 | ENUM | 1 | ADC Mode                          | ,<br>DiffadcLSingadcLDiffadcRSingadcRSingadcLRDiffadcLR                          |
| 19 | ENUM | 1 | ADC MICBIAS Voltage               | VREFx0_8VREFx0_825VREFx0_85VREFx0_875, VREFx0_9VREFx0_925VREFx0_95V<br>REFx0_975 |
| 20 | ENUM | 1 | ADC Main MICBIAS                  | Off, On                                                                          |
| 21 | ENUM | 1 | ADC MIC Left Switch               | , WorkMute                                                                       |
| 22 | ENUM | 1 | ADC MIC Right Switch              | , WorkMute                                                                       |
| 23 | INT  | 1 | DAC LINEOUT Volume                | 26 (range 0->30)                                                                 |
| 24 | INT  | 1 | DAC HPMIX Volume                  | 1 (range 0->2)                                                                   |
| 25 | ENUM | 1 | DAC Control Manually              | , NoneOffOn                                                                      |

There are lots of nodes, and it is going to introduce some commonly used node information here:

|   |      |   |                              |                                   |
|---|------|---|------------------------------|-----------------------------------|
| 0 | ENUM | 1 | I2STDM Digital Loopback Mode | , DisabledMode1Mode2Mode2<br>Swap |
|---|------|---|------------------------------|-----------------------------------|

This node is actually the control node exposed to users by I2S0, but it is closely related to ACodec. It indicates whether the I2STDM controller is working in recovery mode. In the node information:

- Disabled: Default state, recovery mode is disabled
- Mode1: Suitable for 4ch usage scenarios. Channels 1-2 are data of MIC pickup, and channels 3-4 are recovery data.
- Mode2: Suitable for 2ch usage scenarios. The left channel is data of MIC pickup, and the right channel is the playback data of the right channel.
- Mode2 Swap: suitable for 2ch usage scenarios. The left channel is the recovery data of the left channel played, and the right channel is the MIC pickup data. Mode2 Swap and Mode2 channel order are opposite

Therefore, if Mode2 and Mode2 Swap are playing 2ch data, it is best to perform software MIX before sending data to ACodec DAC, so that the recovery reference signal is closer to the signal of the real echo path.

**Remarks: the selected item displayed in tinymix style is added with a comma "," on the left. For example, the displayed ", Disabled" means that the "Disabled" state is currently selected, and so on.**

|   |     |   |                    |                |
|---|-----|---|--------------------|----------------|
| 1 | INT | 1 | ADC MIC Left Gain  | 2 (range 0->3) |
| 2 | INT | 1 | ADC MIC Right Gain | 2 (range 0->3) |

ACodec ADC Boost Gain is the codec analog volume. The actual valid value range is 1 ~ 3:

- vol 0: Disabled and not recommended
- vol 1: 0dB
- vol 2: 20dB
- vol 3: 12dB

|   |     |   |                      |                 |
|---|-----|---|----------------------|-----------------|
| 3 | INT | 1 | ADC ALC Left Volume  | 6 (range 0->31) |
| 4 | INT | 1 | ADC ALC Right Volume | 6 (range 0->31) |

ACodec ADC ALC PGA Gain is the codec analog volume. The value range is 0~31:

- min: -9dB (vol: 0)
- max: +37.5dB (vol: 31)
- step: +1.5dB
- location: 0dB (vol: 6)

|   |     |   |                          |                    |
|---|-----|---|--------------------------|--------------------|
| 5 | INT | 1 | ADC Digital Left Volume  | 195 (range 0->255) |
| 6 | INT | 1 | ADC Digital Right Volume | 195 (range 0->255) |

ACodec ADC Digital Gain is the codec digital volume. The value range is 0~255:

- min: -97.5dB (vol: 0)
- max: +30dB (vol: 255)
- step: +0.5dB
- location: 0dB (vol: 195)

|                                                    |      |   |          |   |
|----------------------------------------------------|------|---|----------|---|
| 18                                                 | ENUM | 1 | ADC Mode | , |
| DiffadclSingadclDiffadcRSingadcRSingadclRDiffadclR |      |   |          |   |

"Diff" is the abbreviation of "Differential"; "Sing" is the abbreviation of "Single-end". Used to configure the ACodec ADC to work in differential or single-ended mode. The default is "Diff" differential mode. And in order to minimize power consumption to the greatest extent, the ADC only enables the L left channel. Therefore, "Diffadcl" is the default preference.

**It should be noted that due to different chip orientation, RV1103 does not support two-channel pickup in differential mode, that is, it does not support the "DiffadclR" option.**

|    |     |   |                    |                  |
|----|-----|---|--------------------|------------------|
| 23 | INT | 1 | DAC LINEOUT Volume | 26 (range 0->30) |
|----|-----|---|--------------------|------------------|

ACodec DAC Lineout Gain, the volume value range is 0~30:

- min: -39dB (vol: 0)
- max: +6dB (vol: 30)

- step: +1.5dB
- location: 0dB (vol: 26)

|    |     |   |                  |                |
|----|-----|---|------------------|----------------|
| 24 | INT | 1 | DAC HPMIX Volume | 1 (range 0->2) |
|----|-----|---|------------------|----------------|

ACodec DAC HPMIX Gain, which is the preamp Gain of Lineout, is usually not set. The value range of the volume value is actually 1 and 2:

- vol 0: Disabled and not recommended
- vol 1: 0dB
- vol 2: 6dB

|    |      |   |                      |             |
|----|------|---|----------------------|-------------|
| 25 | ENUM | 1 | DAC Control Manually | , NoneOffOn |
|----|------|---|----------------------|-------------|

Developers usually use the media framework with the AO Stream is always on (always enabled the sound card). When stopping playback, they only need to quickly turn off and on the peripheral PA. Therefore, a "DAC Control Manually" node is added here:

- None: Default status, the media framework should be actually closed and AO Stream be enabled to control the sound card's off and on.
- Off: When the AO Stream is always on, you can just manually turn off the peripheral PA to make the speaker playback silent.
- On: When the AO Stream is always on and the previous state is Off, you can manually turn on the peripheral PA again to resume playback of the speaker.

The above are `codec control` node that are commonly used by developers during the development process. For other options, you can refer to the `control name` and options and configure and adjust them according to the specific project.

## 3.4 RV1106B/RV1103B ACodec User Level Configuration

We can also use `rk_mpi_amix_test` to list the control nodes related to RV1106B/RV1103B ACodec:

```
# rk_mpi_amix_test --list_contents
cmd parse result:
sound control id      : 0
control name          : (null)
control value         : (null)
list controls         : 0
list contents         : 1
Number of controls: 54
```

| ctl | type | num | name                         | value                      |
|-----|------|-----|------------------------------|----------------------------|
| 0   | ENUM | 1   | SAI Transmit Start Mode Sel  | , StandaloneChained        |
| 1   | ENUM | 1   | SAI Receive Start Mode Sel   | , StandaloneChained        |
| 2   | ENUM | 1   | SAI Transmit SD0x Select     | , AutoSD0x1SD0x2SD0x3SD0x4 |
| 3   | ENUM | 1   | SAI Receive SDIx Select      | , AutoSDIx1SDIx2SDIx3SDIx4 |
| 4   | INT  | 1   | SAI Receive Mono Slot Select | 0 (range 0->128)           |
| 5   | ENUM | 1   | SAI Receive Mono Switch      | , DisableEnable            |

|                                           |      |   |                                      |                                               |
|-------------------------------------------|------|---|--------------------------------------|-----------------------------------------------|
| 6                                         | ENUM | 1 | SAI Transmit Mono Switch             | , DisableEnable                               |
| 7                                         | ENUM | 1 | SAI SDI3 Loopback I2S LR Channel Sel | , L:MIC R:LPL:LP R:MIC                        |
| 8                                         | ENUM | 1 | SAI SDI2 Loopback I2S LR Channel Sel | , L:MIC R:LPL:LP R:MIC                        |
| 9                                         | ENUM | 1 | SAI SDI1 Loopback I2S LR Channel Sel | , L:MIC R:LPL:LP R:MIC                        |
| 10                                        | ENUM | 1 | SAI SDI0 Loopback I2S LR Channel Sel | , L:MIC R:LPL:LP R:MIC                        |
| 11                                        | ENUM | 1 | SAI SDI3 Loopback I2S LR Switch      | , DisableEnable                               |
| 12                                        | ENUM | 1 | SAI SDI2 Loopback I2S LR Switch      | , DisableEnable                               |
| 13                                        | ENUM | 1 | SAI SDI1 Loopback I2S LR Switch      | , DisableEnable                               |
| 14                                        | ENUM | 1 | SAI SDI0 Loopback I2S LR Switch      | , DisableEnable                               |
| 15                                        | ENUM | 1 | SAI SDI3 Loopback Src Select         | , From SD00From SD01From<br>SD02From SD03     |
| 16                                        | ENUM | 1 | SAI SDI2 Loopback Src Select         | , From SD00From SD01From<br>SD02From SD03     |
| 17                                        | ENUM | 1 | SAI SDI1 Loopback Src Select         | , From SD00From SD01From<br>SD02From SD03     |
| 18                                        | ENUM | 1 | SAI SDI0 Loopback Src Select         | , From SD00From SD01From<br>SD02From SD03     |
| 19                                        | ENUM | 1 | SAI SDI3 Loopback Switch             | , DisableEnable                               |
| 20                                        | ENUM | 1 | SAI SDI2 Loopback Switch             | , DisableEnable                               |
| 21                                        | ENUM | 1 | SAI SDI1 Loopback Switch             | , DisableEnable                               |
| 22                                        | ENUM | 1 | SAI SDI0 Loopback Switch             | , DisableEnable                               |
| 23                                        | ENUM | 1 | SAI Sync Out Switch                  | , From CRUFrom IO                             |
| 24                                        | ENUM | 1 | SAI Sync In Switch                   | , From IOFrom Sync Port                       |
| 25                                        | ENUM | 1 | SAI Receive PATH3 Source Select      | From SDI0From SDI1From SDI2,<br>From SDI3     |
| 26                                        | ENUM | 1 | SAI Receive PATH2 Source Select      | From SDI0From SDI1, From<br>SDI2From SDI3     |
| 27                                        | ENUM | 1 | SAI Receive PATH1 Source Select      | From SDI0, From SDI1From<br>SDI2From SDI3     |
| 28                                        | ENUM | 1 | SAI Receive PATH0 Source Select      | , From SDI0From SDI1From<br>SDI2From SDI3     |
| 29                                        | ENUM | 1 | SAI Transmit SD03 Source Select      | From PATH0From PATH1From<br>PATH2, From PATH3 |
| 30                                        | ENUM | 1 | SAI Transmit SD02 Source Select      | From PATH0From PATH1, From<br>PATH2From PATH3 |
| 31                                        | ENUM | 1 | SAI Transmit SD01 Source Select      | From PATH0, From PATH1From<br>PATH2From PATH3 |
| 32                                        | ENUM | 1 | SAI Transmit SD00 Source Select      | , From PATH0From PATH1From<br>PATH2From PATH3 |
| 33                                        | BOOL | 1 | SAI Clk Auto Switch                  | Off                                           |
| 34                                        | INT  | 1 | SAI PCM Read Wait Time MS            | 0 (range 0->10000)                            |
| 35                                        | INT  | 1 | SAI PCM Write Wait Time MS           | 0 (range 0->10000)                            |
| 36                                        | INT  | 1 | ADC MIC Left Gain                    | 2 (range 0->3)                                |
| 37                                        | INT  | 1 | ADC ALC Left Volume                  | 6 (range 0->31)                               |
| 38                                        | INT  | 1 | ADC Digital Left Volume              | 195 (range 0->255)                            |
| 39                                        | ENUM | 1 | ADC HPF Cut-off                      | , OffOn                                       |
| 40                                        | INT  | 1 | ALC AGC Left Volume                  | 12 (range 0->31)                              |
| 41                                        | INT  | 1 | ALC AGC Left Max Volume              | 7 (range 0->7)                                |
| 42                                        | INT  | 1 | ALC AGC Left Min Volume              | 0 (range 0->7)                                |
| 43                                        | ENUM | 1 | ALC AGC Left Switch                  | , OffOn                                       |
| 44                                        | ENUM | 1 | AGC Left Approximate Sample Rate     | ,                                             |
| 96KHz48KHz44.1KHz32KHz24KHz16KHz12KHz8KHz |      |   |                                      |                                               |

```

45      ENUM      1      ADC Mode                                ,
DiffadcLSingadcLDiffadcRSingadcRSingadcLRDiffadcLR
46      ENUM      1      ADC MICBIAS Voltage
VREFx0_8VREFx0_825VREFx0_85VREFx0_875, VREFx0_9VREFx0_925VREFx0_95V
REFx0_975
47      ENUM      1      ADC Main MICBIAS                        Off, On
48      ENUM      1      ADC MIC Left Switch                      , WorkMute
49      INT       1      DAC LINEOUT Volume                       26 (range 0->30)
50      INT       1      DAC HPMIX Volume                         1 (range 0->2)
51      ENUM      1      DAC Control Manually                     , NoneOffOn
52      ENUM      1      DAC VCM Manually                         , NoneOffOn
53      ENUM      1      Audio Mux Select                          ,
Path0Path1Path2Path3Path4Path5

```

As you can see, because the RV1106B/RV1103B ACodec has only one ADC input, the ADC "Right" part is removed compared to the RV1106/RV1103, leaving only the "Left" part, and the functional configuration is exactly the same as that of the RV1106/RV1103. A new "Audio Mux Select" function is added for the SAI controller to connect the internal ACodec and external IO channel switching function:

```

53      ENUM      1      Audio Mux Select                          ,
Path0Path1Path2Path3Path4Path5

```

There are 6 Path control states for channel switching:

- Path0: Play: Output only from ACodec Lineout; Record: Input only from ACodec Mic (default)
- Path1: Play: Output only from ACodec Lineout; Record: Input only from SAI SDI IO
- Path2: Play: Output from ACodec Lineout and SAI SDO IO at the same time; Record: Input only from ACodec Mic
- Path3: Play: Output from ACodec Lineout and SAI SDO IO at the same time; Record: Input only from SAI SDI IO
- Path4: Play: Output only from SAI SDO IO; Record: Input only from ACodec Mic
- Path5: Play: Output only from SAI SDO IO; Record: Input only from SAI SDI IO (for testing only, usually not used)

In addition, due to the upgrade of the controller SAI, the original RV1106/RV1103 The I2S part is quite different from the current SAI, but we mainly focus on the configuration related to the RV1106B/RV1103B digital loopback. Because the RV1103B/RV1106B ACodec has only one ADC input, the Data Lane related to the SAI input is only the SDI0 part:

```

10      ENUM      1      SAI SDI0 Loopback I2S LR Channel Sel    , L:MIC R:LPL:LP R:MIC

```

- L:MIC R:LP: SAI SDI0 Lane left and right channel selection for loopback mode: left channel is MIC, right channel is loopback
- L:LP R:MIC: SAI SDI0 Lane left and right channel selection for loopback mode: right channel is MIC, left channel is loopback

```

14      ENUM      1      SAI SDI0 Loopback I2S LR Switch        , DisableEnable

```

SAI SDI0 Lane left and right channel selection mode switch of the acquisition mode is disabled by default and remains in data pass-through mode. Enable it when digital acquisition is required.

|    |      |   |                              |                                        |
|----|------|---|------------------------------|----------------------------------------|
| 18 | ENUM | 1 | SAI SDI0 Loopback Src Select | , From SD00From SD01From SD02From SD03 |
|----|------|---|------------------------------|----------------------------------------|

Indicates the selection of the reference output data Lane after SAI SDI0 starts digital acquisition. Since RV1106B/RV1103B ACodec only supports mono recording and mono playback, only SDO0 is supported as a reference here. Keep the default state.

|    |      |   |                          |                 |
|----|------|---|--------------------------|-----------------|
| 22 | ENUM | 1 | SAI SDI0 Loopback Switch | , DisableEnable |
|----|------|---|--------------------------|-----------------|

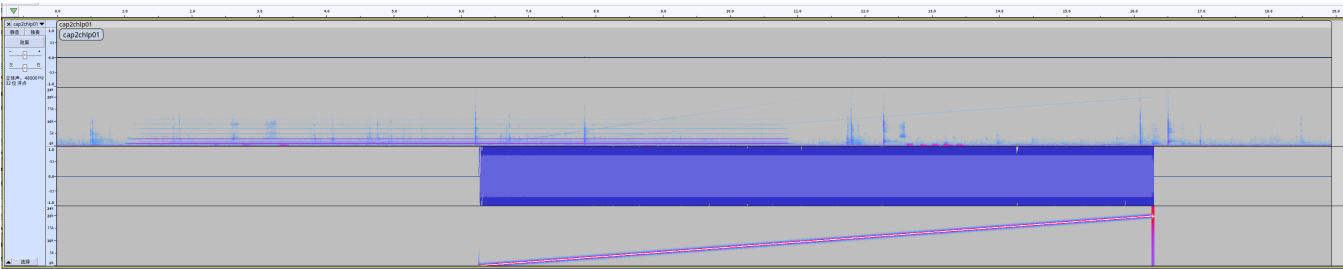
Digital retrieving switch of SAI SDI0 Lane. It is off by default and can be turned on when retrieving is needed.  
In summary, if you want to fully configure single MIC single retrieving, you can configure it according to the following command:

```
rk_mpi_amix_test --control "SAI SDI0 Loopback I2S LR Channel Sel" --value "L:MIC R:LP"
rk_mpi_amix_test --control "SAI SDI0 Loopback Src Select" --value "From SD00"
rk_mpi_amix_test --control "SAI SDI0 Loopback I2S LR Switch" --value "Enable"
rk_mpi_amix_test --control "SAI SDI0 Loopback Switch" --value "Enable"
```

Since the "SAI SDI0 Loopback I2S LR Channel Sel" and "SAI SDI0 Loopback Src Select" registers maintain the required state after power-on, the configuration command for enabling digital acquisition can usually be omitted as follows:

```
rk_mpi_amix_test --control "SAI SDI0 Loopback I2S LR Switch" --value "Enable"
rk_mpi_amix_test --control "SAI SDI0 Loopback Switch" --value "Enable"
```

RV1106B/RV1103B 2-channel capture example, the left channel is MIC, and the right channel is loopback:



### 3.5 RV1126B/RV1126BP ACodec User-Level Configuration

Currently, the latest SDK kernel code enables ADC0 and ADC1 by default:



```
commit b69407b9ad1232f35ae108fd8755d962fcdeda84
Author: Xing Zheng <zhengxing@rock-chips.com>
Date: Tue Jul 1 18:44:24 2025 +0800
```

```
arm64: dts: rockchip: rv1126b-evb: enable both acodec_pmu and acodec for 4ch capture by default
```

You can refer to the following patch through the engineering code to confirm whether it is enabled:

```
diff --git a/arch/arm64/boot/dts/rockchip/rv1126b-evb2-v10.dtsi
b/arch/arm64/boot/dts/rockchip/rv1126b-evb2-v10.dtsi
index 1ca72d41c5ab..cf6b5788ea5e 100644
--- a/arch/arm64/boot/dts/rockchip/rv1126b-evb2-v10.dtsi
+++ b/arch/arm64/boot/dts/rockchip/rv1126b-evb2-v10.dtsi
@@ -445,7 +445,6 @@ &rtc {
};

&sai2 {
-   rockchip,sai-rx-route = <1 0 2 3>;
   status = "okay";
   /delete-property/ pinctrl-names;
   /delete-property/ pinctrl-0;

diff --git a/arch/arm64/boot/dts/rockchip/rv1126b-evb.dtsi
b/arch/arm64/boot/dts/rockchip/rv1126b-evb.dtsi
index eb3879ca4c5f..8ca9b5b9a32c 100644
--- a/arch/arm64/boot/dts/rockchip/rv1126b-evb.dtsi
+++ b/arch/arm64/boot/dts/rockchip/rv1126b-evb.dtsi
@@ -20,7 +20,7 @@ &acodec_sound: acodec-sound {
    rockchip,format = "i2s";
    rockchip,mclk-fs-mapping = <256 1024>;
    rockchip,cpu = <&sai2>;
-   rockchip,codec = <&audio_codec>, <&acdcdig_dsm>;
+   rockchip,codec = <&audio_codec_pmu>, <&audio_codec>, <&acdcdig_dsm>;
};

adc-keys {

diff --git a/arch/arm64/boot/dts/rockchip/rv1126b-evb1-v10.dtsi
b/arch/arm64/boot/dts/rockchip/rv1126b-evb1-v10.dtsi
index 31faaa3ad88b..da82560304cd 100644
--- a/arch/arm64/boot/dts/rockchip/rv1126b-evb1-v10.dtsi
+++ b/arch/arm64/boot/dts/rockchip/rv1126b-evb1-v10.dtsi
@@ -132,6 +132,10 @@ &acodec_sound {
    status = "okay";
};

+&audio_codec_pmu {
+   status = "okay";
+};
+
&audio_codec {
    status = "okay";
```

```
};
```

Therefore, the contents of a default sound card typically look like this:

```
# rk_mpi_amix_test --list_contents 1
cmd parse result:
sound control id      : 0
control name          : (null)
control value         : (null)
list controls         : 0
list contents         : 1
ctl    type    num    name                                value
0      BOOL    1      Headphone Switch                    1
1      BOOL    1      ACodec ADC Switch                    0
2      INTE    1      ACodec Digital Gain Volume          0(0->127)
3      ENUM    1      ACodec HPF Cutoff                    , "3.79Hz" "60Hz" "243Hz" "493Hz"
4      BOOL    1      ACodec HPF Switch                    0
5      INTE    1      ACodec PGA Gain Volume              16(0->31)
6      BOOL    1      ACodec_LP ADC Switch                    0
7      INTE    1      ACodec_LP Digital Gain Volume        0(0->127)
8      ENUM    1      ACodec_LP HPF Cutoff                    , "3.79Hz" "60Hz" "243Hz" "493Hz"
9      BOOL    1      ACodec_LP HPF Switch                    0
10     INTE    1      ACodec_LP PGA Gain Volume            16(0->31)
11     INTE    2      DAC Digital Volume                    255(0->510) 255(0->510)
12     ENUM    1      DAC HPF Cutoff                        , "80Hz" "100Hz" "120Hz" "140Hz"
13     BOOL    1      DAC HPF Switch                    0
14     BOOL    1      Headset Mic Switch                    1
15     BOOL    1      Main Mic Switch                      1
16     ENUM    1      Power Amplifier                      , "Off" "On"
17     BOOL    1      SAI2 Clk Auto Switch                    0
18     INTE    1      SAI2 Receive Mono Slot Select        0(0->128)
19     ENUM    1      SAI2 Receive Mono Switch              , "Disable" "Enable"
20     ENUM    1      SAI2 Receive PATH0 Source Sele        , "From SDI0" "From SDI1" "From SDI2"
"From SDI3"
21     ENUM    1      SAI2 Receive PATH1 Source Sele        "From SDI0" , "From SDI1" "From SDI2"
"From SDI3"
22     ENUM    1      SAI2 Receive PATH2 Source Sele        "From SDI0" "From SDI1" , "From SDI2"
"From SDI3"
23     ENUM    1      SAI2 Receive PATH3 Source Sele        "From SDI0" "From SDI1" "From SDI2"
, "From SDI3"
24     ENUM    1      SAI2 Receive SDIx Select                , "Auto" "SDIx1" "SDIx2" "SDIx3"
"SDIx4"
25     ENUM    1      SAI2 Receive Start Mode Sel            , "Standalone" "Chained"
26     ENUM    1      SAI2 SDI0 Loopback I2S LR Chan          , "L:MIC R:LP" "L:LP R:MIC"
27     ENUM    1      SAI2 SDI0 Loopback I2S LR Swit          , "Disable" "Enable"
28     ENUM    1      SAI2 SDI0 Loopback Src Select          , "From SD00" "From SD01" "From SD02"
"From SD03"
29     ENUM    1      SAI2 SDI0 Loopback Switch              , "Disable" "Enable"
30     ENUM    1      SAI2 SDI1 Loopback I2S LR Chan          , "L:MIC R:LP" "L:LP R:MIC"
31     ENUM    1      SAI2 SDI1 Loopback I2S LR Swit          , "Disable" "Enable"
32     ENUM    1      SAI2 SDI1 Loopback Src Select          , "From SD00" "From SD01" "From SD02"
"From SD03"
33     ENUM    1      SAI2 SDI1 Loopback Switch              , "Disable" "Enable"
```

|    |      |   |                                |                                                         |
|----|------|---|--------------------------------|---------------------------------------------------------|
| 34 | ENUM | 1 | SAI2 SDI2 Loopback I2S LR Chan | ,"L:MIC R:LP" "L:LP R:MIC"                              |
| 35 | ENUM | 1 | SAI2 SDI2 Loopback I2S LR Swit | ,"Disable" "Enable"                                     |
| 36 | ENUM | 1 | SAI2 SDI2 Loopback Src Select  | ,"From SD00" "From SD01" "From SD02"<br>"From SD03"     |
| 37 | ENUM | 1 | SAI2 SDI2 Loopback Switch      | ,"Disable" "Enable"                                     |
| 38 | ENUM | 1 | SAI2 SDI3 Loopback I2S LR Chan | ,"L:MIC R:LP" "L:LP R:MIC"                              |
| 39 | ENUM | 1 | SAI2 SDI3 Loopback I2S LR Swit | ,"Disable" "Enable"                                     |
| 40 | ENUM | 1 | SAI2 SDI3 Loopback Src Select  | ,"From SD00" "From SD01" "From SD02"<br>"From SD03"     |
| 41 | ENUM | 1 | SAI2 SDI3 Loopback Switch      | ,"Disable" "Enable"                                     |
| 42 | ENUM | 1 | SAI2 Sync In Switch            | ,"From IO" "From Sync Port"                             |
| 43 | ENUM | 1 | SAI2 Sync Out Switch           | ,"From CRU" "From IO"                                   |
| 44 | ENUM | 1 | SAI2 Transmit Mono Switch      | ,"Disable" "Enable"                                     |
| 45 | ENUM | 1 | SAI2 Transmit SD00 Source Sele | ,"From PATH0" "From PATH1" "From<br>PATH2" "From PATH3" |
| 46 | ENUM | 1 | SAI2 Transmit SD01 Source Sele | "From PATH0" ,"From PATH1" "From<br>PATH2" "From PATH3" |
| 47 | ENUM | 1 | SAI2 Transmit SD02 Source Sele | "From PATH0" "From PATH1" ,"From<br>PATH2" "From PATH3" |
| 48 | ENUM | 1 | SAI2 Transmit SD03 Source Sele | "From PATH0" "From PATH1" "From<br>PATH2" ,"From PATH3" |
| 49 | ENUM | 1 | SAI2 Transmit SDOx Select      | ,"Auto" "SD0x1" "SD0x2" "SD0x3"<br>"SD0x4"              |
| 50 | ENUM | 1 | SAI2 Transmit Start Mode Sel   | ,"Standalone" "Chained"                                 |
| 51 | BOOL | 1 | Speaker Switch                 | 1                                                       |
| 52 | BOOL | 1 | hp switch                      | 0                                                       |
| 53 | BOOL | 1 | spk switch                     | 0                                                       |
| 54 | INTE | 1 | SAI2 PCM Read Wait Time MS     | 0(0->10000)                                             |
| 55 | INTE | 1 | SAI2 PCM Write Wait Time MS    | 0(0->10000)                                             |

Because the RV1126B's ADC0 and ADC1 each occupy an I2S data line, SDI0 and SDI1, respectively, each SDI line receives two identical channels of data during recording. For example, the left and right channels on SDI0 are connected to MIC0 data from ADC0, and the left and right channels on SDI1 are connected to MIC1 data from ADC1.

Therefore, whether using a single or dual microphone, it's recommended to enable the capture function on both SDI0 and SDI1:

```
rk_mpi_amix_test --control 'SAI2 SDI0 Loopback I2S LR Switch' --value 'Enable'
rk_mpi_amix_test --control 'SAI2 SDI0 Loopback Switch' --value 'Enable'
rk_mpi_amix_test --control 'SAI2 SDI1 Loopback I2S LR Switch' --value 'Enable'
rk_mpi_amix_test --control 'SAI2 SDI1 Loopback Switch' --value 'Enable'
```

This way, the digital audio data played back via SAI2 SDO0 by default can be directly obtained from the right channel of SDI0 and SDI1.

For a common 2-channel mic + 1-channel audio capture scenario, the RV1126B uses the full alsalib architecture by default and supports asound.conf configuration (note that asound.conf must be placed in the /etc directory):

```
# Copyright (c) 2025 Rockchip Electronics Co. Ltd.
# Author: Xing Zheng <zhengxing@rock-chips.com>
```

```
pcm.!default {
    type asym
    playback.pcm "play_plug"
    capture.pcm "capt_softvol"
}

ctl.!default {
    type hw
    card 0
}

# ----- Capture -----
pcm.cap_multi {
    type multi
    slaves.a.pcm "hw:0,0"
    slaves.a.channels 4
    bindings.0.slave a
    bindings.0.channel 0
    bindings.1.slave a
    bindings.1.channel 2
    bindings.2.slave a
    bindings.2.channel 1
}

pcm.capt_plug {
    type plug
    slave.pcm "cap_multi"
}

pcm.capt_softvol {
    type softvol
    slave.pcm "capt_plug"
    control.name "Master Capture Volume"
    control.card 0
    min_dB -51.0
    max_dB 0.0
    resolution 101
}

# ----- Playback -----
pcm.play_softvol {
    type softvol
    slave.pcm "hw:0,0"
    control.name "Master Playback Volume"
    control.card 0
    min_dB -51.0
    max_dB 0.0
    resolution 101
}

pcm.play_plug {
    type plug
```

```

slave.pcm "play_softvol"
}

```

Next, refer to test\_mpi\_ai.cpp on the application side and modify it as follows to adjust the layout bitmaps of Rec 2ch and Ref 1ch respectively during 3-channel recording:

```

diff --git a/src/tests/rt_mpi/mod/test_mpi_ai.cpp b/src/tests/rt_mpi/mod/test_mpi_ai.cpp
index 682467c03..798907c0d 100644
--- a/src/tests/rt_mpi/mod/test_mpi_ai.cpp
+++ b/src/tests/rt_mpi/mod/test_mpi_ai.cpp
@@ -757,9 +760,13 @@ RK_S32 test_init_ai_vqe(TEST_AI_CTX_S *params) {
    stAiVqeConfig.s32WorkSampleRate = params->s32SampleRate;
    stAiVqeConfig.s32FrameSample = params->s32SampleRate * params->s32VqeGapMs / 1000;

+   /* FIXME: for capture with more channel */
    if (params->s32DeviceChannel == 4) {
        stAiVqeConfig.s64RefChannelType = 0x0c;
        stAiVqeConfig.s64RecChannelType = 0x03;
+   } else if (params->s32DeviceChannel == 3) {
+       stAiVqeConfig.s64RefChannelType = 0x04;
+       stAiVqeConfig.s64RecChannelType = 0x03;
    } else {
        stAiVqeConfig.s64RefChannelType = 0x02;
        stAiVqeConfig.s64RecChannelType = 0x01;

```

Use rk\_mpi\_ai\_test 3ch recording, and specify the entry node as "default" for sound\_card\_name:

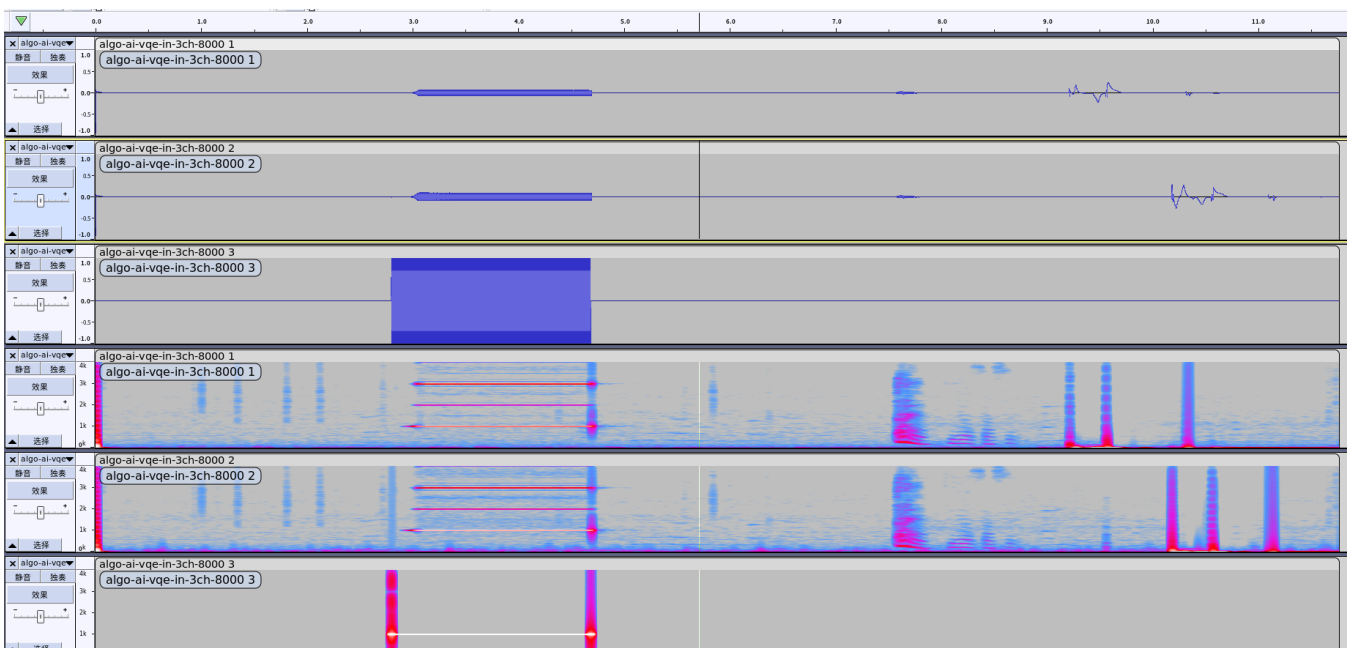
```

export rt_dump_skv_inout=true

rk_mpi_ai_test --sound_card_name=default --device_rate 8000 --device_ch 3 --out_rate 8000 --
out_ch 1 --output /tmp --vqe_enable 1 --vqe_cfg /tmp/config_aivqe.json

```

The expected arrangement of MIC0/MIC1/Loopback0 data is as follows. The following diagrams are the time and frequency domain diagrams of these three channels:



## 4. ACodec Commonly Used Test Commands

Since RV1106/RV1103/RV1106B/RV1103B operating environment is usually based on the rockit framework, the following are combination of basic examples and rockit related commands.

### 4.1 Commonly Used Playback Test Commands

The following uses a PCM with a sampling rate of 16kHz/2ch/16bit as the playback sound source. The rk\_mpi\_ao\_test command handles 16bit depth by default. in which, the parameters "device\_rate=16000" and "input\_rate=16000" are the same, indicating that the re-sampling function does not need to be enabled; the parameter "input\_ch=2" needs to correspond to the original sound source 2ch:

```
rk_mpi_ao_test -i /tmp/sine_16k_500_1000_0dbfs.wav --sound_card_name=hw:0,0 --device_ch=2 --device_rate=16000 --input_rate=16000 --input_ch=2
```

The sine\_16k\_500\_1000\_0dbfs.wav file can be downloaded from: <https://redmine.rock-chips.com/attachments/2297743>.

For more command parameter usage, please refer to the "MPI" related documents in the SDK Rockit project directory and the relevant test cases in the mpi example directory.

### 4.2 Commonly Used Capture Test Commands

The following uses PCM format recording with a sampling rate of 16kHz/2ch/16bit. The rk\_mpi\_ai\_test command also processes 16 bit depth by default. in which, the parameters "device\_rate=16000" and "input\_rate=16000" are the same, indicating that the re-sampling function does not need to be enabled; the parameters "device\_ch=2" and "input\_ch=2" are the same, indicating that channel processing is not required. The data generated by recording is the cap\_out.pcm file in the /tmp directory.

```
rk_mpi_ai_test --sound_card_name=hw:0,0 --device_rate=16000 --device_ch=2 --out_rate=16000 --out_ch=2 --output=/tmp
```

For more command parameter usage, please refer to the "AO/AI debugging steps" chapter in "Rockchip\_Developer\_Guide\_MPI\_FAQ.pdf".